

ACHISAY, Kazakhstan

WMO: 381960

Lat: 43.550N

Lon: 68.900E

Elev: 2697

StdP: 13.32

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.2	9.4	-10.9	3.4	14.2	-5.0	4.6	17.4	36.3	10.1	34.1	9.3	6.2	320	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	94.0	61.9	91.2	61.1	88.7	60.2	63.8	86.6	62.7	85.6	61.7	84.3	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.8	75.9	68.3	54.7	70.4	67.9	52.8	65.5	67.7	30.5	86.3	29.6	85.5	28.9	84.4	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	12.8	10.0	DB	0.8	99.0	5.7	2.2	-3.3	100.6	-6.7	101.9	-9.9	103.1	-14.1	104.8	(4)
(5)			WB	0.0	66.8	5.6	1.8	-4.0	68.1	-7.3	69.1	-10.5	70.1	-14.6	71.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	28.8	29.7	42.3	52.8	64.3	73.7	78.2	75.8	65.4	52.0	39.8	30.8	(6)
(7)		DBStd	19.74	9.01	10.23	9.70	9.28	7.33	6.39	6.24	6.70	8.26	9.55	10.34	9.87	(7)
(8)		HDD50	2599	659	569	273	76	3	0	0	0	3	90	329	597	(8)
(9)		HDD65	5633	1123	988	705	376	104	9	2	5	96	410	755	1060	(9)
(10)		CDD50	3668	1	2	33	159	447	711	874	801	463	152	24	1	(10)
(11)		CDD65	1228	0	0	0	9	83	270	411	341	107	7	0	0	(11)
(12)		CDH74	12090	0	0	1	50	545	2650	4617	3432	757	38	0	0	(12)
(13)		CDH80	4643	0	0	0	4	103	958	1998	1360	215	5	0	0	(13)
(14)	Wind	WSAvg	4.7	4.0	4.4	4.4	5.4	5.3	4.7	5.0	5.9	5.5	4.2	3.7	3.7	(14)
(15)	Precipitation	PrecAvg	15.6	1.9	1.9	2.2	1.9	1.5	0.5	0.5	0.2	0.2	0.9	1.9	2.2	(15)
(16)		PrecMax	28.8	4.3	4.3	4.5	5.6	4.9	1.9	2.1	0.7	1.1	2.7	6.3	10.3	(16)
(17)		PrecMin	6.2	0.0	0.0	0.6	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	5.0	1.0	1.1	1.3	1.5	1.2	0.6	0.6	0.2	0.3	0.9	1.6	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	58.3	69.5	79.2	86.5	94.4	98.6	96.8	89.9	78.8	68.7	56.0	(19)
(20)			MCWB	41.4	46.4	52.2	58.5	59.6	62.5	63.4	62.4	59.1	55.2	52.1	42.6	(20)
(21)		2%	DB	47.2	51.1	63.9	74.4	82.8	91.1	94.8	93.2	85.5	73.1	62.2	50.3	(21)
(22)			MCWB	38.6	42.9	49.9	56.7	58.4	61.3	62.3	61.4	57.3	51.9	48.0	39.9	(22)
(23)		5%	DB	43.5	46.5	59.4	70.3	79.7	88.4	91.7	90.1	82.0	69.0	57.7	46.7	(23)
(24)			MCWB	36.1	39.7	47.7	54.3	57.1	60.5	61.4	60.4	56.0	50.3	46.4	37.7	(24)
(25)		10%	DB	40.2	43.0	55.6	66.4	76.8	85.5	89.2	87.2	78.2	65.0	53.4	43.5	(25)
(26)			MCWB	33.9	37.4	46.4	52.8	56.0	59.6	60.7	59.3	54.6	48.7	44.6	36.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	47.2	54.9	61.2	61.9	65.3	65.7	64.0	60.3	56.8	53.0	44.9	(27)
(28)			MCD B	50.4	54.4	61.3	73.4	79.5	84.0	89.3	88.8	87.3	73.8	63.2	52.6	(28)
(29)		2%	WB	39.6	44.3	51.9	58.4	60.4	63.4	64.0	62.6	58.1	54.0	51.0	41.6	(29)
(30)			MCD B	45.4	49.8	61.1	69.1	77.4	84.5	87.2	88.4	82.0	68.4	59.3	46.7	(30)
(31)		5%	WB	37.4	40.9	49.5	56.3	59.0	62.2	63.0	61.5	56.7	52.0	48.0	39.2	(31)
(32)			MCD B	41.8	45.2	58.0	67.4	74.7	83.0	86.3	86.9	79.7	65.5	56.4	45.2	(32)
(33)		10%	WB	35.3	37.9	47.5	54.2	57.7	61.1	62.0	60.3	55.1	49.9	45.0	36.8	(33)
(34)			MCD B	39.4	42.4	54.5	64.5	72.7	81.6	85.2	84.9	76.8	63.1	52.1	43.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.1	12.0	13.4	13.9	15.8	17.0	16.1	15.3	14.6	14.7	13.0	12.2	(35)
(36)			MCDBR	14.0	15.3	18.5	18.6	18.6	19.3	17.3	17.8	18.1	19.1	17.3	15.0	(36)
(37)			MCWBR	9.4	9.7	10.3	9.6	8.0	7.3	6.5	6.9	7.8	9.3	9.6	9.5	(37)
(38)		5% WB	MCD BR	11.8	13.7	16.4	15.7	17.6	18.3	17.4	17.1	17.7	17.5	15.4	13.6	(38)
(39)			MCWBR	9.0	9.7	10.5	9.1	8.4	7.6	7.0	7.1	8.2	9.1	9.4	9.5	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.331	0.367	0.424	0.453	0.441	0.425	0.413	0.379	0.356	0.381	0.374	0.327	(41)	
(42)		taud	2.057	2.031	1.987	1.971	2.055	2.132	2.176	2.263	2.286	2.122	2.081	2.091	(42)	
(43)		Ebn at Noon	239	253	253	257	265	269	271	277	275	245	221	228	(43)	
		Edh at Noon	34	42	50	55	52	48	46	40	36	37	33	30		
(44)	All-Sky Solar Radiation	RadAvg	515	826	1201	1710	2183	2425	2379	2157	1690	1072	613	431	(44)	
(45)		RadStd	73	87	92	167	129	103	104	69	69	81	63	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+1.75	N/A	N/A	N/A	N/A	+329	+239

Nomenclature: See separate page